BIG DATA ANALYTICS / WEEK 3

Machine Learning & Experimental Evaluation

M5 Sales Forecasting
MySQL → KNIME → MySQL → Power BI



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Week 2 Recap & Week 3 Goals

A / DATA AND DATABASE

Joined M5 sales history with the calendar and prepared SQL for the sales_predictions table.

B / KNIME

Week 2 reported 87,390 rows after cleaning. The updated workflow shows Linear Regression training and prediction.

C / POWER BI

The dashboard draft shows Actual vs Predicted. The remaining handoff is to verify that Power BI reads the predictions written by KNIME to MySQL.

Feature Selection & Experimental Design

DATA PREPARATION

SQL joins sales history to the calendar and derives the is_holiday flag.

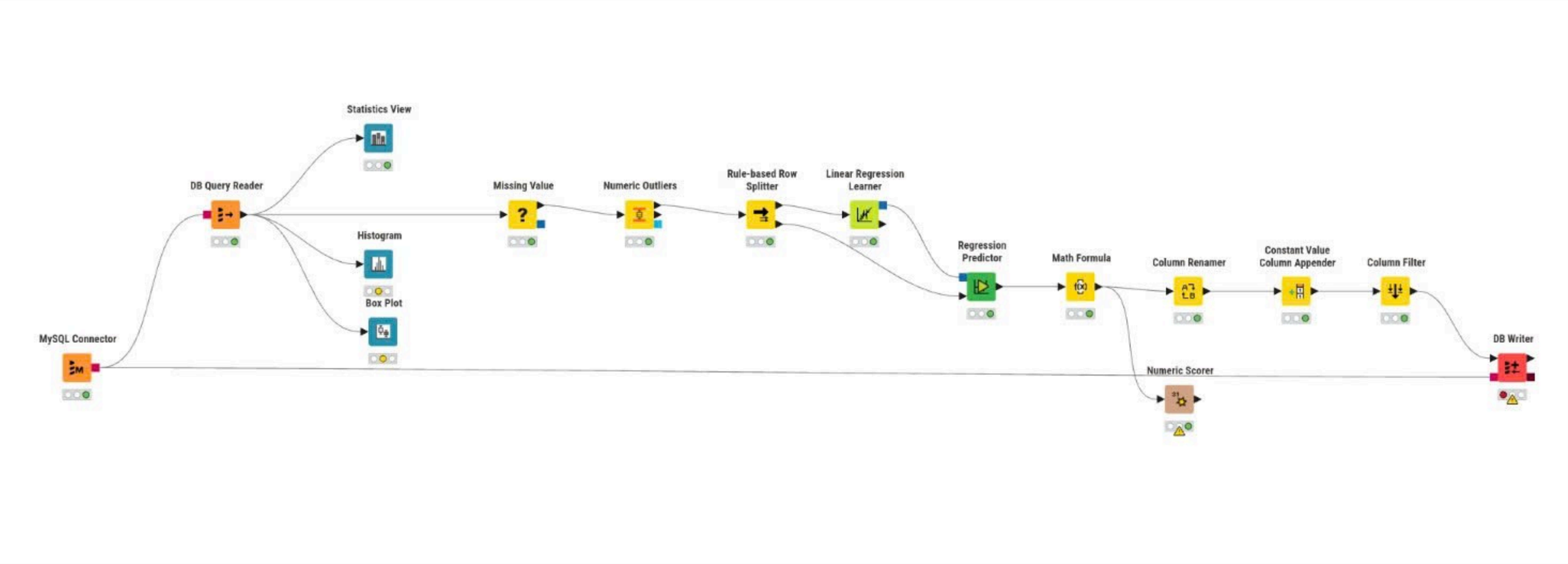
TARGET AND SPLIT

The model predicts sales. KNIME uses a Rule-based Row Splitter before training and prediction.

DETAILS TO CONFIRM

Record the final model inputs and train/test cutoff. Verify how missing values and outliers were handled without using test information.

Linear Regression Workflow in KNIME



The workflow reads from MySQL, prepares the data, splits the rows, trains the model and generates predictions.

Model Evaluation & Business Interpretation

EVALUATION STEP

The workflow sends predictions to Numeric Scorer. The screenshot shows a warning on the scorer.

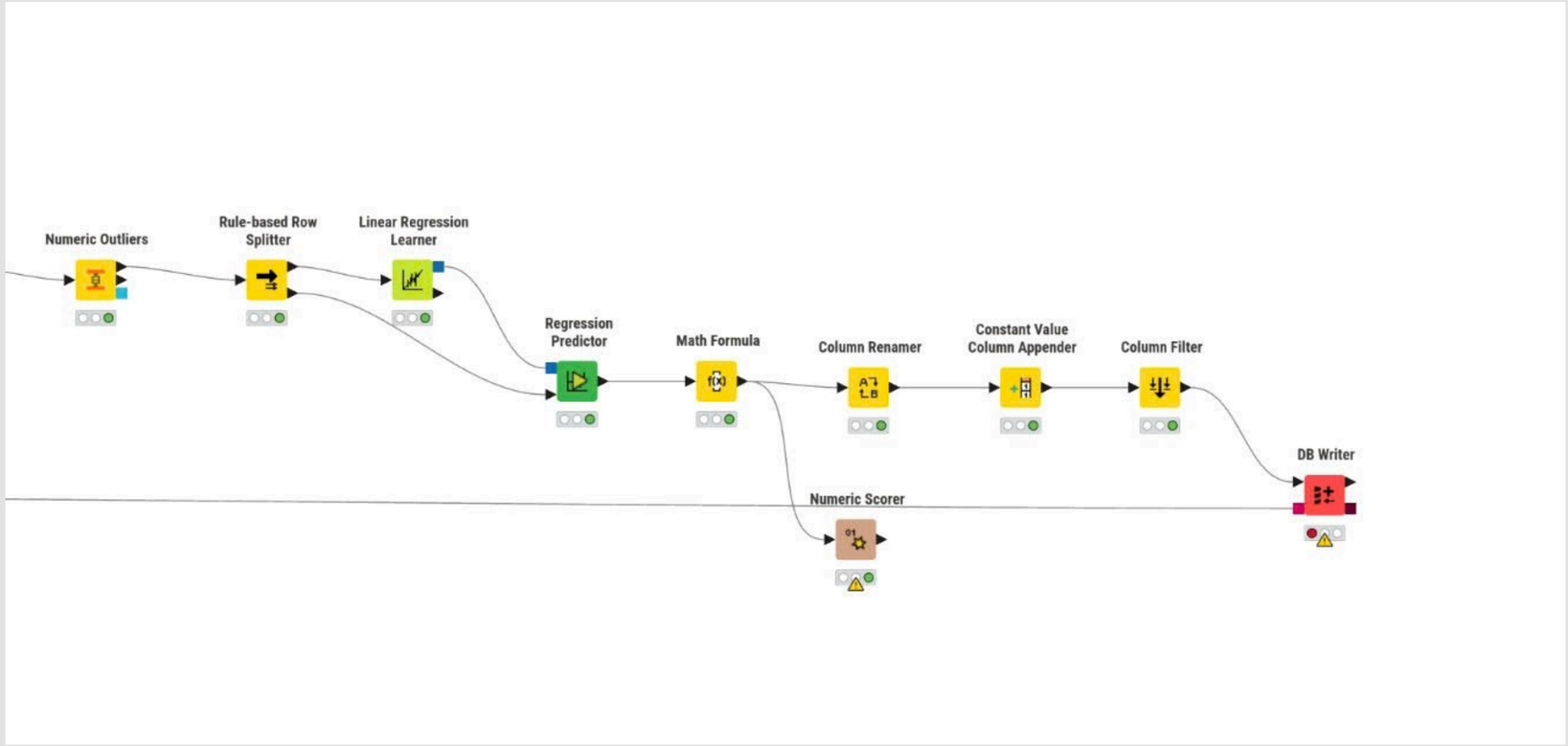
RESULTS TO CONFIRM

Review the warning and record the test-set MAE, RMSE or R^2 from the scorer output.

BUSINESS INTERPRETATION

Use the measured error to explain how reliable the forecasts are for the team’s decision questions.

Writing Predictions Back to MySQL



The SQL for the receiving table is ready. DB Writer shows a red status in the supplied screenshot.

MySQL Verification

WRITE STATUS

Verify that DB Writer completes successfully.

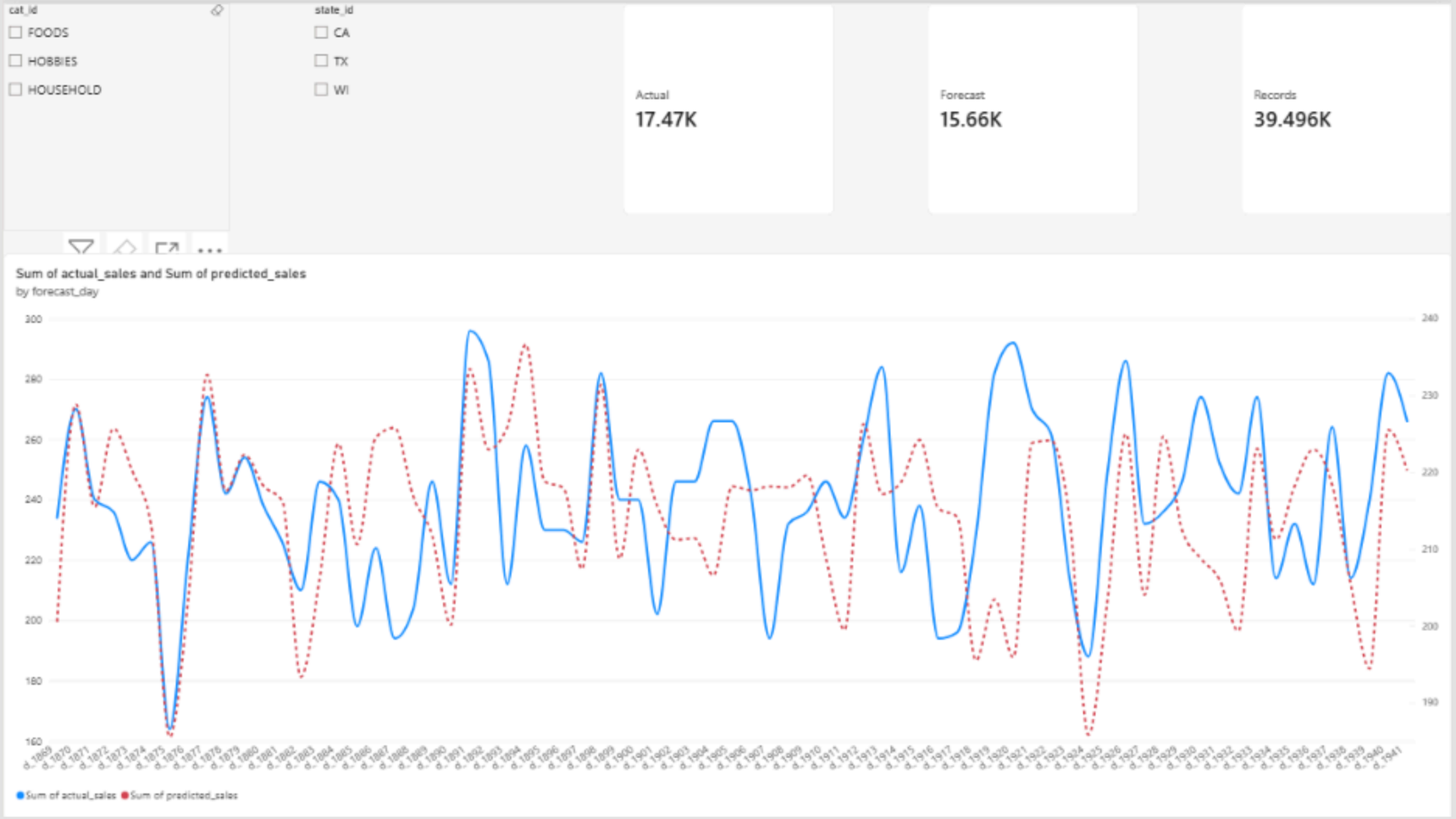
RESULTING TABLE

Check the row count and show sample predictions in MySQL.

POWER BI HANDOFF

Confirm that Power BI reads the predictions written by KNIME to MySQL.

Initial Power BI Dashboard



Actual vs Predicted by forecast day

Confirm the MySQL source and demonstrate the filters.

Dashboard KPIs & Filters

KPIs

Actual, Forecast and Records.

FILTERS

Category and state.

MODEL RESULT

Actual vs Predicted by forecast day.

Research Paper Progress

WEEK 2 REPORT

The Week 2 slides report completed Introduction, Dataset and Data Quality sections.

IN PROGRESS

Architecture and EDA were in progress.

WEEK 3 DRAFT

Document the method and experimental results with verified test metrics and supporting tables or figures.

Week 4 Plan

A AND B

Verify the MySQL write-back.

C

Verify the dashboard source.

TEAM

Update the work breakdown and planned versus actual hours.

Week 3 Delivery Summary

WHAT

Demonstrate the KNIME sales prediction workflow and the Power BI draft. Show the MySQL receiving table and verify the prediction write-back.

WHY

Use Linear Regression for a numeric sales target. Store predictions in MySQL so Power BI can compare actual and predicted sales by day, category and state.

WHEN

Week 3 review covers the workflow and dashboard draft. The remaining verification and report updates are tracked in the Week 4 plan.

HOW MUCH

Week 2 slides planned 7.0 hours for Week 3. Actual Week 3 hours and the gap are pending the team’s time log.